01 Fuel Pump Relay Output
02 Idle valve close control
03 Injector 5 (M50TU): injector 1 (M60/S50/M50)
04 Injector 6 (M50TU): injector 4 (M60): injector 3 (S50/M50)
05 Injector 4 (M50TU): injector 6 (M60): injector 2 (S50/M50)
06 Power/Chassis GND ground (Injector/output/ignition return)
07 VANOS solenoid (M50TU) / injector 7 (M60)
08 Check Engine Light Output
09 Unspecified OEM terminal; independent pass-through
10 Unspecified OEM terminal; independent pass-through
11 Throttle-position output to transmission / traction control (variant dependent)
12 Universal analog input. Suggested TPS 1 sensor input (M50); oxygen sensor 2 input (M60/S50)
13 Oxygen sensor 1 input (M50TU/M60/S50) / MAF burn-off (M50)
14 GNDA Analog/Sensor Ground (MAF return; JP14 to ECU GNDA)
15 Ignition monitor / return (do not assume chassis ground)
16 Crank sensor input (M50TU/M60/S50) / cam sensor negative (M50)
17 Cam sensor input (M50TU/M60/S50) / fuel-consumption output (M50)
18 Auxiliary circuit (not specified for M3x in reference)
19 Secondary air pump control (M60/S50: verify equipment)
20 Unspecified OEM terminal; independent pass-through
21 Unspecified OEM terminal; independent pass-through
22 Ignition primary output, driver channel 7 (M60: coil 7)
23 Ignition primary output, driver channel 4 (M50TU: coil 4; M60: coil 6; S50: coil 2; M50: coil 2)
24 Ignition primary output, driver channel 6 (M50TU: coil 6; M60: coil 4; S50: coil 3; M50: coil 3)
25 Ignition primary output, driver channel 5 (M50TU: coil 5; M60: coil 1; S50: coil 1; M50: coil 1)
26 12V Battery Supply (constant, terminal 30)
27 Main Relay Control Output
28 Power/Chassis GND ground (electronics/shields: connected to chassis in M50TU/M60 wiring: JP28 …
29 Idle valve open control
30 Unspecified OEM terminal; independent pass-through

31 Injector 3 (M50TU): injector 5 (M60/S50/M50)
32 Injector 2 (M50TU): injector 8 (M60): injector 6 (S50/M50)
33 Injector 1 (M50TU): injector 3 (M60): injector 4 (S50/M50)
34 Power/Chassis GND ground (Injector/output/ignition return)
35 Injector 2 output (M60)
36 EVAP canister purge valve control
37 Oxygen sensor heater relay control (M60/S50/M50: separate from pin 38)
38 Oxygen sensor heater relay control (M50TU: separate from pin 37)
39 GNDA Analog/Sensor Ground (oxygen sensor 2 return, M60/S50: JP39 to ECU GNDA)
40 GNDA Analog/Sensor Ground (oxygen sensor 1 return, M50TU/M60/S50: JP40 to ECU GNDA)
41 MAF Sensor Input
42 Vehicle speed input (M50TU/M60/S50)
43 Crank sensor return (M50TU/M60/S50); sensor ground on M50 reference
44 Cam sensor return (M50TU/M60/S50) / cam sensor positive (M50)
45 Ignition-monitor shield / return (verify harness)
46 Fuel consumption output (TI)
47 Tachometer Output (TD; M50TU/M60/S50)
48 A/C compressor relay control
49 Ignition primary output, driver channel 8 (M60: coil 2)
50 Ignition primary output, driver channel 1 (M50TU: coil 1; M60: coil 3; S50: coil 4; M50: coil 4)
51 Ignition primary output, driver channel 2 (M50TU: coil 2; M60: coil 8; S50: coil 6; M50: coil 6)
52 Ignition primary output, driver channel 3 (M50TU: coil 3; M60: coil 5; S50: coil 5; M50: coil 5)
53 Unspecified OEM terminal; independent pass-through
54 12V supply from main relay (terminal 87)
55 Power/Chassis GND ground (Injector/output/ignition return)
56 Ignition Switch / Battery Voltage Analog Input (terminal 15)
57 Transmission Ignition-timing Intervention (M50TU)
58 Unspecified OEM terminal; independent pass-through
59 5V Sensor Supply (TPS)
60 Diagnostic programming-voltage circuit (verify protocol)

61 Unspecified OEM terminal; independent pass-through
62 Transmission / traction-control signal (variant dependent)
63 Unspecified OEM terminal; independent pass-through
64 A/C request (M50TU/M60/S50) / transmission timing intervention (M50)
65 A/C pressure input (M50TU/M60/S50) / transmission P/N (M50)
66 Immobilizer / drive-away protection input (variant dependent)
67 Knock sensor, cylinders 7-8 (M60) / crank positive (M50)
68 Knock sensor, cylinders 5-6 (M60/S50) / crank positive (M50)
69 Knock sensor, cylinders 4-6 (M50TU) or 3-4 (M60/S50)
70 Knock sensor, cylinders 1-3 (M50TU) or 1-2 (M60/S50) / oxygen signal (M50)
71 Knock-sensor shield/return (M50TU/M60/S50) / oxygen sensor return (M50); JP71 selects GND or G…
72 Unspecified OEM terminal; independent pass-through
73 Universal analog input. Suggested TPS 1 sensor input (M50TU/M60/S50); VSS on M50
74 Tachometer output (M50 reference; otherwise unspecified)
75 Unspecified OEM terminal; independent pass-through
76 Unspecified OEM terminal; independent pass-through
77 IAT Sensor input
78 ECT CLT Coolant Sensor Input
79 Unspecified OEM terminal; independent pass-through
80 Unspecified OEM terminal; independent pass-through
81 Transmission P/N (M50TU); headlights-on signal (M60); verify variant
82 ABS/traction-control interface (verify harness)
83 ABS ignition intervention (verify harness)
84 Shield / return (M60/S50); independent contact
85 CAN bus Low (M60/S50) / A/C pressure input (M50)
86 CAN bus High (M60/S50) / auxiliary fan signal (M50)
87 Diagnostic RXD circuit
88 Diagnostic TXD circuit

BMW 88 / Superseal 60+34 — eight ignition power stages
Scope: Bosch M3.1/M3.3/M3.3.1, NOT Siemens MS40/MS41. Cylinder order differs by engine.
J3 pins 6/34/55 are power GND. All other OEM pins are independent. All JPs are OPEN.
GNDA is ECU sensor ground on 30C. JP14/39/40 join sensor returns to GNDA.
JP28/45/84 join chassis/shields to GND. JP71: 1=GNDA, 2=OEM71, 3=GND; bridge one side only.
Do not close links on signal variants. Pins 15/44 are deliberately not ground-jumpered.
Q1–Q8: 22 ohm series gate resistance; 100k input pulldown, same topology as Ford adapter.
Use 5 V ignition commands and ECU-controlled dwell. Channel number is not cylinder number.
No dwell timeout/current limiter. Load and thermal validation required for assembled hardware.
A=J1.1…34, B=J1.35…60, C=J2.1…34. Pinouts: rusefi.com/docs/pinouts/bmw88/ and /bmw88-adapter/.

rusEFI
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File: bmw88-superseal.kicad_sch
Title: BMW Motronic 88 to Superseal 60+34
Size: A2 | Date: 2026-10-07 | Rev: a
KiCad E.D.A. 10.0.6 | Id: 1/1